

Azure Data Factory — Data Flow Walkthrough

Step-by-step guide: building a Mapping Data Flow that reads `weather_data.csv` from Blob Storage, converts temperature from Celsius to Fahrenheit, selects the final columns, and writes the result to a new CSV file — then running it via a pipeline.

Overview of the process

1. Create a Dataflow and add a Blob Storage source (`weather_data.csv`).
2. Enable Data Flow Debug and preview the raw source data.
3. Add a Derived Column transformation to calculate $\text{temp_F} = (\text{temperature_c} * 1.8) + 32$.
4. Add a Select transformation to keep only the required columns.
5. Add a Sink transformation to write the results to a new CSV dataset in the 'output' container.
6. Create a pipeline with a Data Flow activity referencing the dataflow.
7. Run the pipeline in Debug mode and verify it succeeds.
8. Confirm the output CSV file was created in Blob Storage with the correct data.

1

Create a new Dataflow

In Azure Data Factory (datafactory3354), a new dataflow named **dataflow1** was created under Factory Resources > Data flows. The canvas is empty, showing the "Add Source" placeholder.

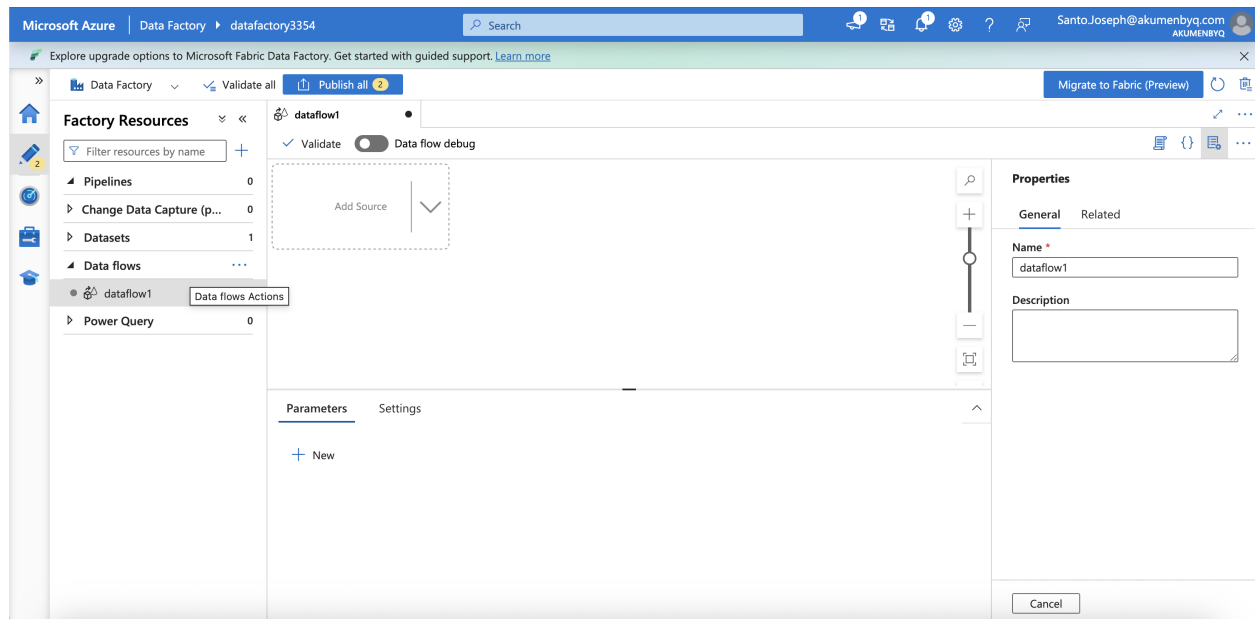


Figure 1: Create a new Dataflow

Add a Source and create a Linked Service

Clicking "Add Source" created **source1**. A new linked service to **Azure Blob Storage** was configured using "Account key" authentication, selecting the subscription and storage account (storage3354).

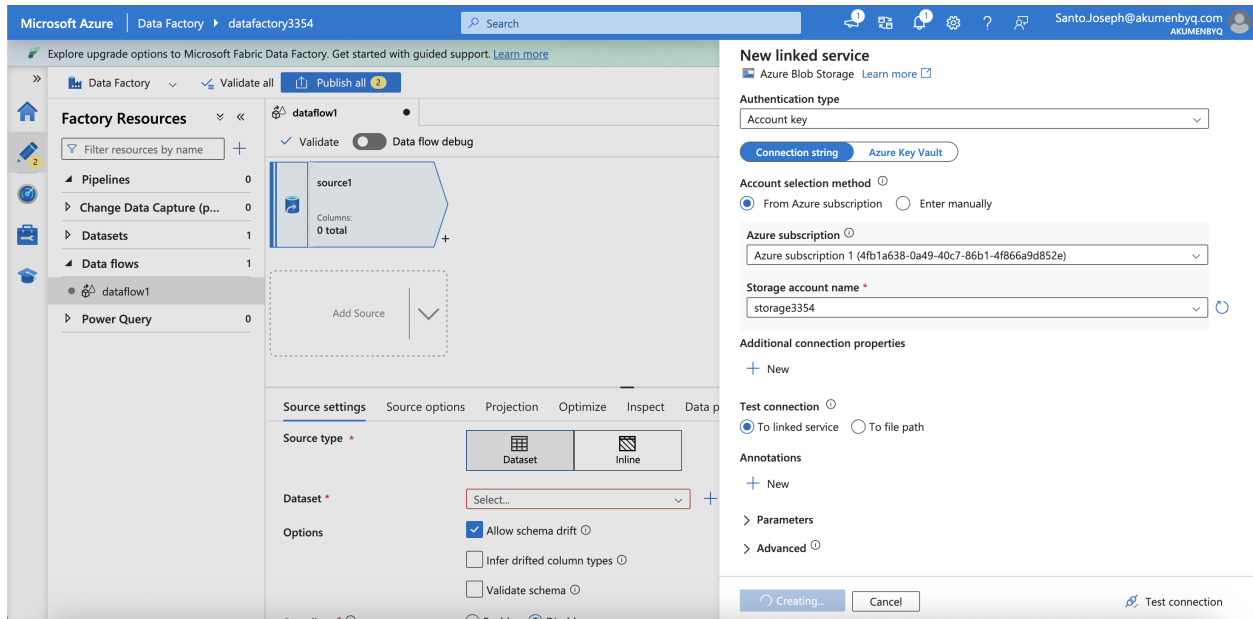


Figure 2: Add a Source and create a Linked Service

Browse and select the source file

Using the Browse dialog, the **input** folder was opened and **weather_data.csv** was selected as the source file (other files temp_1.csv and temp_2.csv were visible but not used).

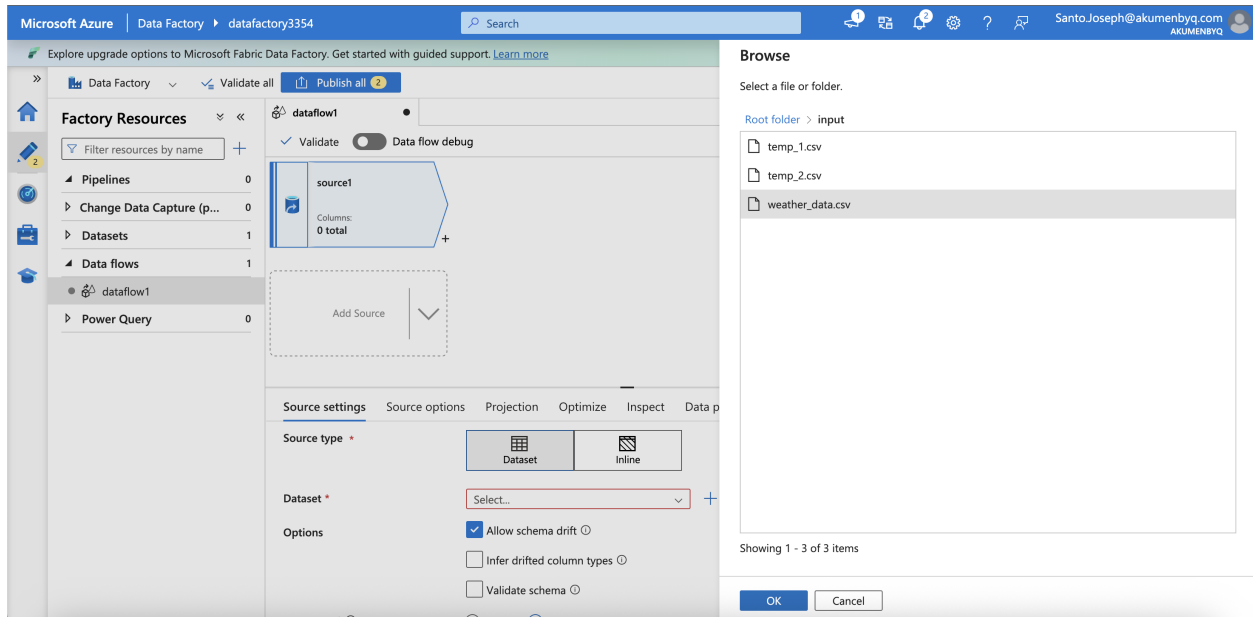


Figure 3: Browse and select the source file

Enable Data Flow Debug and preview data

Data flow debug was turned on. Once the cluster was ready (session started), the source dataset showed 5 columns total, ready for preview.

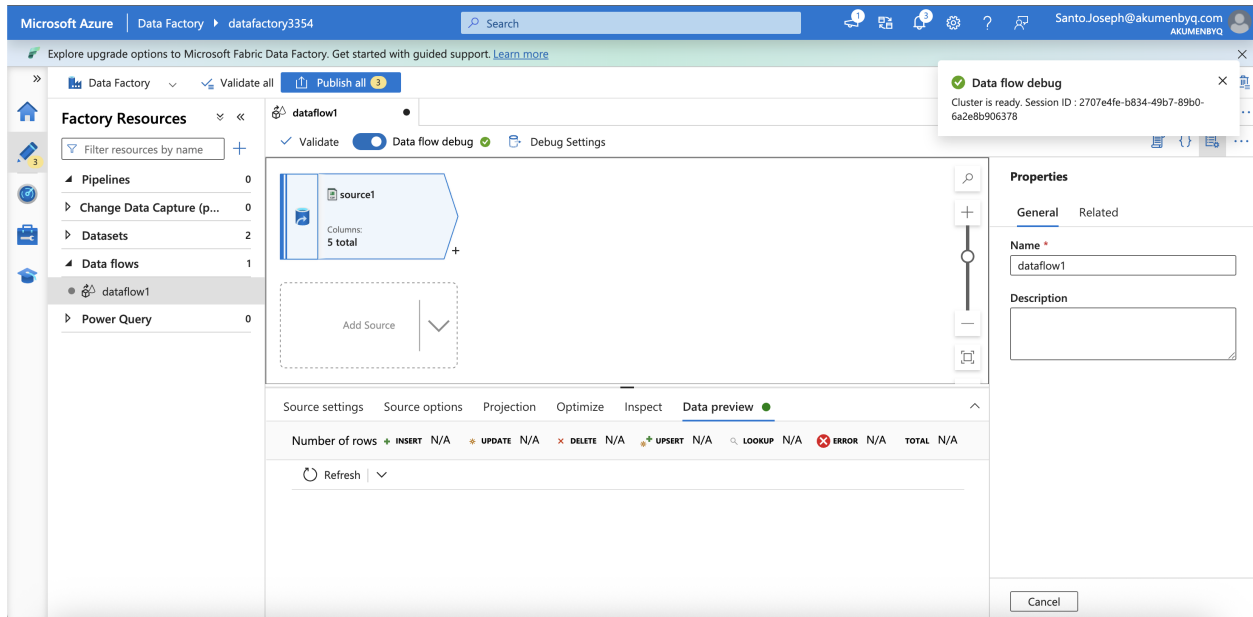


Figure 4: Enable Data Flow Debug and preview data

Preview the source data

The Data Preview tab was refreshed to display the raw contents of `weather_data.csv`: columns **date**, **city**, **temperature_c**, **humidity_pct**, **condition** with 5 sample rows (New York, London, Tokyo, Sydney, Mumbai).

The screenshot shows the Microsoft Azure Data Factory interface. The left sidebar lists 'Factory Resources' including Pipelines, Datasets, and Data flows. The main area shows a data flow named 'dataflow1' with a 'Data preview' tab selected. The preview displays a table with 5 rows of weather data. The right sidebar shows the 'Properties' panel for the data flow.

	date	city	temperature_c	humidity_pct	condition
+	2026-08-19	New York	22.5	60	Sunny
+	2026-08-20	London	18.3	75	Cloudy
+	2026-08-21	Tokyo	25.1	68	Rainy
+	2026-08-22	Sydney	19.8	55	Clear
+	2026-08-23	Mumbai	30.2	80	Humid

Figure 5: Preview the source data

Add a Derived Column transformation

From the "+" menu on source1, the **Schema modifier** > **Derived Column** transformation was selected to compute a new column based on existing ones.

The screenshot displays the Microsoft Azure Data Factory portal for a data factory named 'datafactory3354'. The left sidebar shows the 'Factory Resources' tree with 'Data flows' expanded, highlighting 'dataflow1'. The main canvas shows a data flow diagram with a source named 'source1' (5 total columns) and a 'Derived Column' transformation. The 'Schema modifier' menu is open, showing options like 'Join', 'Conditional Split', 'Exists', 'Union', 'Lookup', 'Select', 'Aggregate', 'Surrogate Key', 'Pivot', and 'Unpivot'. The 'Derived Column' option is selected. The 'Data preview' pane shows a table with 5 columns: 'date', 'abc', 'cit', 'condition', and 'abc'. The 'condition' column contains values: 'Cloudy', 'Clear', 'Sunny', 'Rainy', and 'Humid'. The 'Properties' pane on the right shows the 'General' tab with the name 'dataflow1' and a description field.

date	abc	cit	condition	abc
2026-08-20	Lo		Cloudy	
2026-08-22	Sy		Clear	
2026-08-19	Ne		Sunny	
2026-08-21	To		Rainy	
2026-08-23	Mi		Humid	

Figure 6: Add a Derived Column transformation

Configure the Derived Column settings

The new transformation **derivedColumn1** was added with incoming stream set to source1. A new column was added via "Add column".

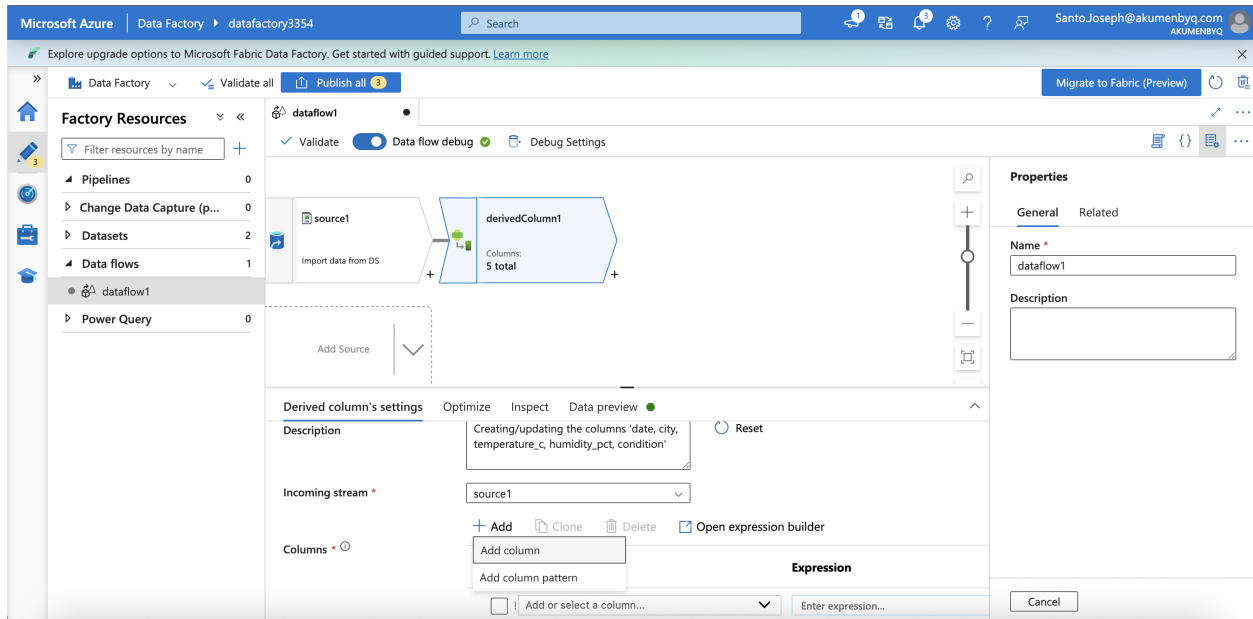


Figure 7: Configure the Derived Column settings

Define the temp_F expression

A new column named **temp_F** was created with the expression:

$(\text{temperature_c} * 1.8) + 32$

This converts the Celsius temperature to Fahrenheit, bringing the total column count to 6.

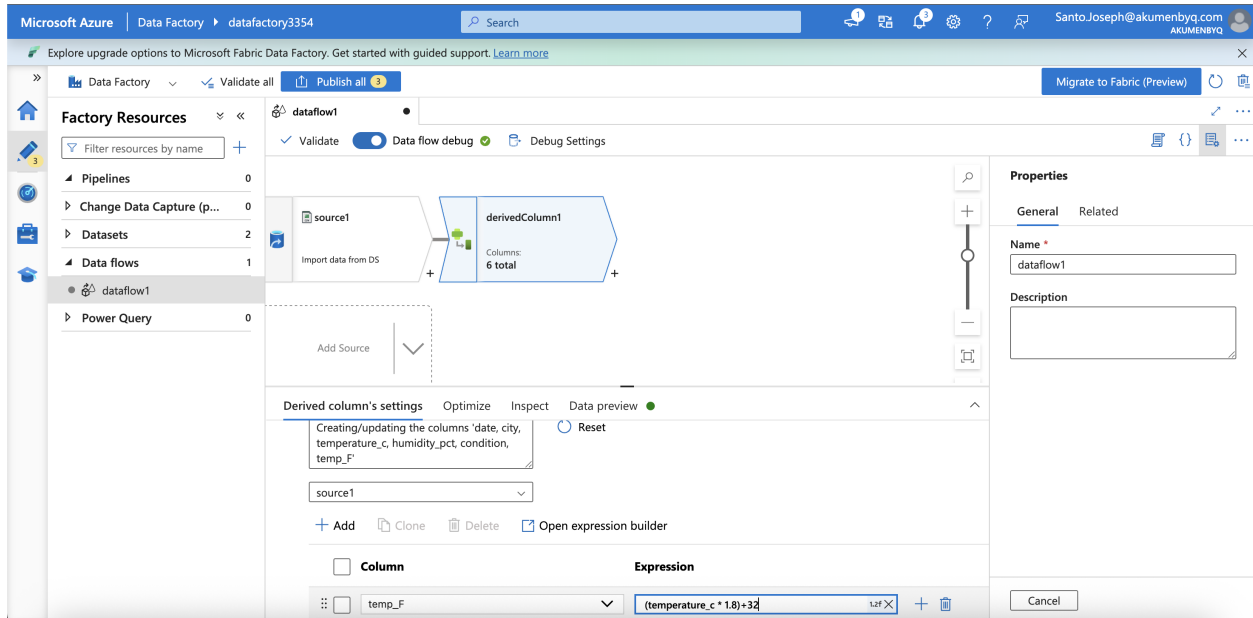


Figure 8: Define the temp_F expression

Verify the Derived Column output

The Data Preview tab confirmed the new **temp_F** column was calculated correctly for every row (e.g. 18°C → 64.4°F, 30°C → 86.0°F).

The screenshot shows the Microsoft Azure Data Factory interface for a data flow named 'dataflow1'. The 'Data preview' tab is selected, displaying a table with 6 rows of data. The columns are: date, city, temperature_c, humidity_pct, condition, and temp_F. The 'temp_F' column shows the calculated values for each row, confirming the conversion from Celsius to Fahrenheit.

date	city	temperature_c	humidity_pct	condition	temp_F
2026-08-20	London	18	75	Cloudy	64.4
2026-08-22	Sydney	19	55	Clear	66.2
2026-08-19	New York	22	60	Sunny	71.6
2026-08-21	Tokyo	25	68	Rainy	77.0
2026-08-23	Mumbai	30	80	Humid	86.0

Figure 9: Verify the Derived Column output

Add a Select transformation

A **Select** transformation (select1) was added after derivedColumn1 to control and rename the output columns, keeping "Skip duplicate input/output columns" enabled.

The screenshot displays the Microsoft Azure Data Factory portal for a data factory named 'datafactory3354'. The left sidebar shows the 'Factory Resources' tree with 'Data flows' expanded, highlighting 'dataflow1'. The main canvas shows a data flow diagram with three components: 'source1' (Import data from DS), 'derivedColumn1' (Reference: 1, Columns: 6 total), and 'select1' (Columns: 6 total). Below the canvas, the 'Select settings' tab is active for the 'select1' transformation. The settings include: 'Output stream name' set to 'select1', a 'Description' field containing 'Renaming derivedColumn1 to select1 with columns 'date, city, temperature_c, humidity_pct, condition, temp_F'', 'Incoming stream' set to 'derivedColumn1', and two checked options: 'Skip duplicate input columns' and 'Skip duplicate output columns'. A 'Reset' button is visible next to the description field. On the right, the 'Properties' pane shows the 'General' tab with 'Name' set to 'dataflow1' and an empty 'Description' field. A 'Cancel' button is at the bottom right of the settings pane.

Figure 10: Add a Select transformation

Map columns in the Select transformation

The Select settings mapped all 5 output columns (date, temperature_c, humidity_pct, condition, temp_F) from derivedColumn1, effectively dropping the unneeded 'city' column.

The screenshot displays the Microsoft Azure Data Factory interface for a data flow named 'dataflow1'. The left sidebar shows the 'Factory Resources' tree with 'Data flows' expanded, highlighting 'dataflow1'. The main canvas shows a data flow diagram with three components: 'source1' (Import data from DS), 'derivedColumn1' (Creating/updating the columns: date, city, temperature_c, humidity_pct, condition, temp_F), and 'select1' (Columns: 5 total). Below the diagram, the 'Select settings' tab is active, showing a table of column mappings. The table has columns for 'derivedColumn1's column', a filter icon, and 'Name as'. Five mappings are listed: 'date' to 'date', 'temperature_c' to 'temperature_c', 'humidity_pct' to 'humidity_pct', 'condition' to 'condition', and 'temp_F' to 'temp_F'. The 'city' column is not mapped. The right sidebar shows the 'Properties' panel for 'dataflow1' with fields for 'Name' and 'Description'. The top navigation bar includes 'Data Factory', 'datafactory3354', a search bar, and user information for 'Santo Joseph@akumenbyq.com'.

derivedColumn1's column	Name as
date	date
temperature_c	temperature_c
humidity_pct	humidity_pct
condition	condition
temp_F	temp_F

Figure 11: Map columns in the Select transformation

12 Preview the Select output

The Data Preview for select1 showed the final column set — date, temperature_c, humidity_pct, condition, temp_F — confirming the transformation logic was correct.

The screenshot displays the Microsoft Azure Data Factory interface for a data flow named 'dataflow1'. The left sidebar shows the 'Factory Resources' tree with 'Data flows' expanded, highlighting 'dataflow1'. The main canvas shows the data flow diagram with three components: 'source1' (Import data from DS), 'derivedColumn1' (Creating/updating the columns: date, city, temperature_c, humidity_pct, condition, temp_F), and 'select1' (Columns: 5 total). Below the diagram, the 'Data preview' tab is active, showing a table of data. The table has columns: date, temperature_c, humidity_pct, condition, and temp_F. The data rows show dates from 2026-08-19 to 2026-08-23 with corresponding temperature, humidity, and condition values. The right sidebar shows the 'Properties' panel for 'dataflow1' with fields for Name and Description.

date	temperature_c	humidity_pct	condition	temp_F
2026-08-19	22	60	Sunny	71.6
2026-08-20	18	75	Cloudy	64.4
2026-08-21	25	68	Rainy	77.0
2026-08-22	19	55	Clear	66.2
2026-08-23	30	80	Humid	86.0

Figure 12: Preview the Select output

13 Add a Sink (Destination)

From the "+" menu on select1, **Sink** was chosen under the Destination section to define where the transformed data would be written.

Microsoft Azure | Data Factory | datafactory3354

Explore upgrade options to Microsoft Fabric Data Factory. Get started with guided support. [Learn more](#)

Data Factory | Validate all | Publish all | Migrate to Fabric (Preview)

Factory Resources | Filter resources by name | Pipelines | Change Data Capture (p...) | Datasets | Data flows | dataflow1 | Power Query

dataflow1 | Validate | Data flow debug | Debug Settings

source1 | derivedColumn1 | select1

Import data from DS | Creating/updating the columns date, city, temperature_c, humidity_pct, condition, temp_f | Columns: 5 total

Add Source

Select settings | Optimize | Inspect | Data preview

Number of rows | INSERT 5 | UPDATE 0 | DELETE 0 | UPSERT 0 | LOOKUP 0

Refresh | Typecast | Modify | Map drifted | Statistics | Remove

date | abc | temperature_c | humidity_pct | condition

2026-08-19 | 22 | 60 | Sunny

2026-08-20 | 18 | 75 | Cloudy

2026-08-21 | 25 | 68 | Rainy

2026-08-22 | 19 | 55 | Clear

Formatters | Flatten | Parse | Stringify | Row modifier | Filter | Sort | Alter Row | Assert | Flowlets | Flowlet | Destination | Sink

Properties | General | Related

Name | dataflow1

Description

Destination for your data flow

Figure 13: Add a Sink (Destination)

14 Configure the Sink dataset

A new dataset (DelimitedText1) was created using linked service **ls_1**, with file path set to the **output** container, first row as header enabled, and schema imported from the connection/store.

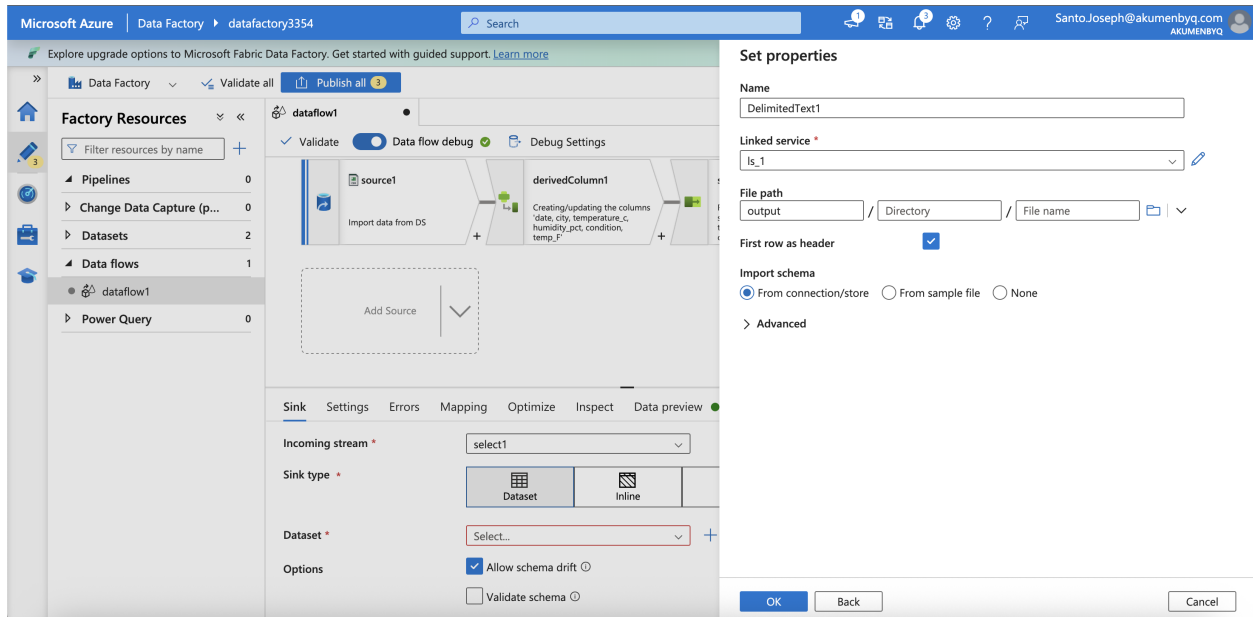


Figure 14: Configure the Sink dataset

15 Verify the Sink connection

The sink (sink1) was linked to the DelimitedText1 dataset. "Test connection" confirmed a **Connection successful** result before publishing.

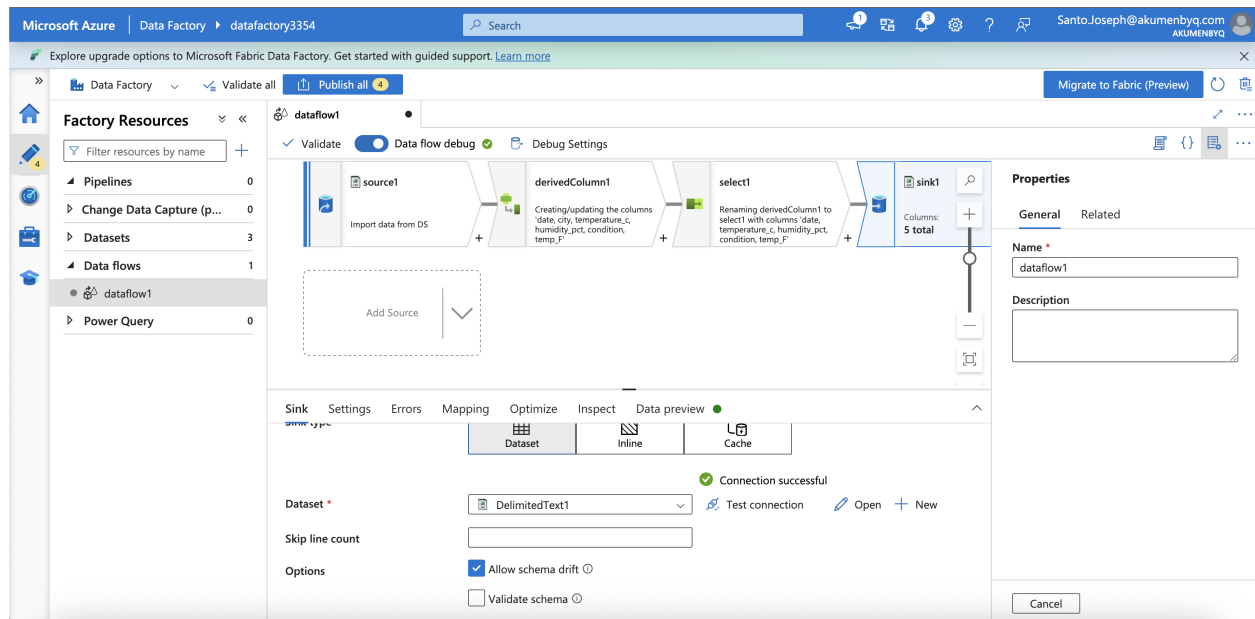


Figure 15: Verify the Sink connection

Create a Pipeline with a Data Flow activity

A new pipeline (**pipeline1**) was created and a **Data flow** activity was dragged onto the canvas, configured to run **dataflow1**.

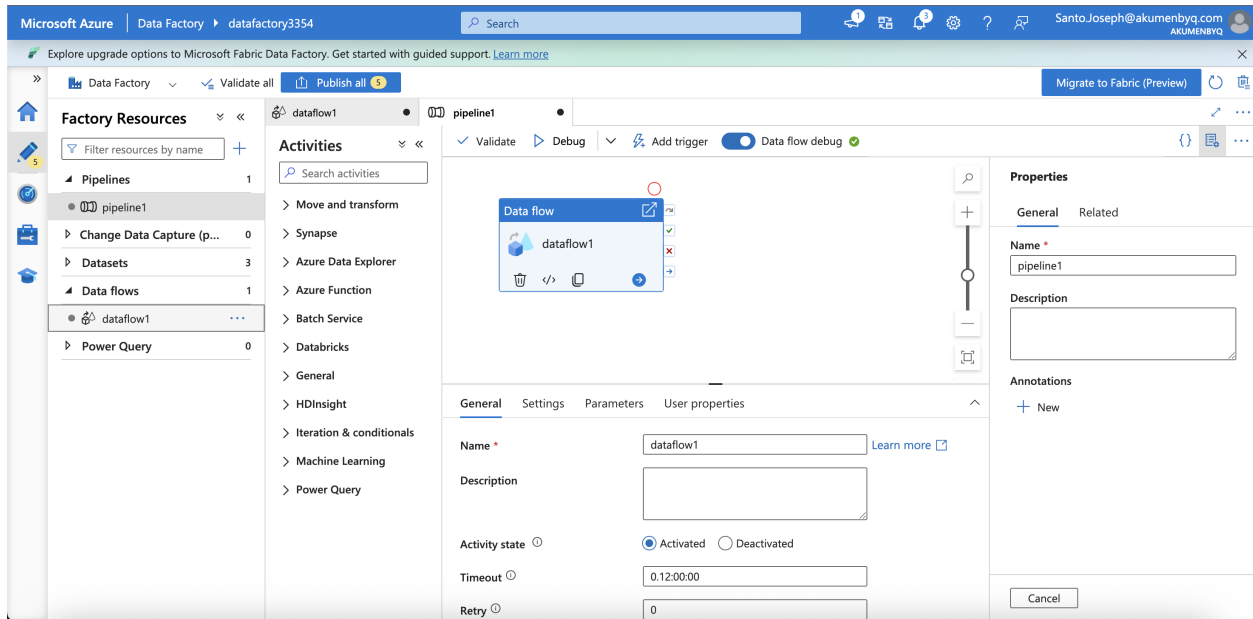


Figure 16: Create a Pipeline with a Data Flow activity

17 Run the pipeline in Debug mode

The pipeline was executed using **Debug**. The Output tab confirmed the run finished with status **Succeeded** for the dataflow1 activity.

The screenshot displays the Microsoft Azure Data Factory interface for a pipeline named 'pipeline1'. The 'Debug' mode is selected, and the 'Data flow debug' toggle is turned on. The pipeline status is 'Succeeded', and the 'dataflow1' activity is also 'Succeeded'. The 'Output' tab is active, showing the pipeline run ID 'e53665fd-1d59-405c-82b3-31623d4f32ce' and the activity status 'Succeeded'.

Factory Resources

- Pipelines: 1
 - pipeline1
- Change Data Capture (p...): 0
- Datasets: 3
- Data flows: 1
 - dataflow1
- Power Query: 0

Activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Properties

General

Name: pipeline1

Description:

Annotations: + New

Output

Pipeline: e53665fd-1d59-405c-82b3-31623d4f32ce

Pipeline status: Succeeded

View debug run consumption

Monitor in Azure Metrics | Export to CSV

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity...	Run start	Run end
dataflow1	Succeeded	Data flow	8/24/2026, 3:38:28 PM	1r

Cancel

Figure 17: Run the pipeline in Debug mode

18 Inspect the debug run output details

Clicking the output icon revealed the detailed run JSON, including the Spark cluster ID, Spark version, and compute acquisition duration used for the debug run.

The screenshot displays the Microsoft Azure Data Factory console for a workspace named 'datafactory3354'. The 'Factory Resources' sidebar on the left shows a list of resources including Pipelines (1), Datasets (3), Data flows (1), and Power Query (0). The 'pipeline1' pipeline is selected, and the 'Data flow debug' tab is active. The 'Activities' pane shows a 'Data flow' activity named 'dataflow1'. The 'Output' pane displays the following JSON:

```
{
  "runStatus": {
    "ClusterId": "dfdebugpool-eastus.debugpool-8Cores-General-70.1985",
    "sparkVersion": "3.4",
    "computeAcquisitionDuration": 1673,
    "version": "20260813.2",
    "profile": {
      "source1": {

```

The 'Properties' pane on the right shows the 'General' tab with the Name 'pipeline1' and a Description field. The 'Annotations' section shows a 'New' annotation. The bottom status bar indicates the 'dataflow1' activity has 'Succeeded' at '8/24/2026, 3:38:28 PM'.

Figure 18: Inspect the debug run output details

Verify the output file in Blob Storage

In the storage account (storage3354), the **output** container now contained a **_SUCCESS** marker file and a generated CSV part-file (190 B), confirming the pipeline wrote data successfully.

Home > resource1 > storage3354 | Containers

output Container

Search

+ Add Directory ↑ Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive) Only show active blobs

Showing all 2 items

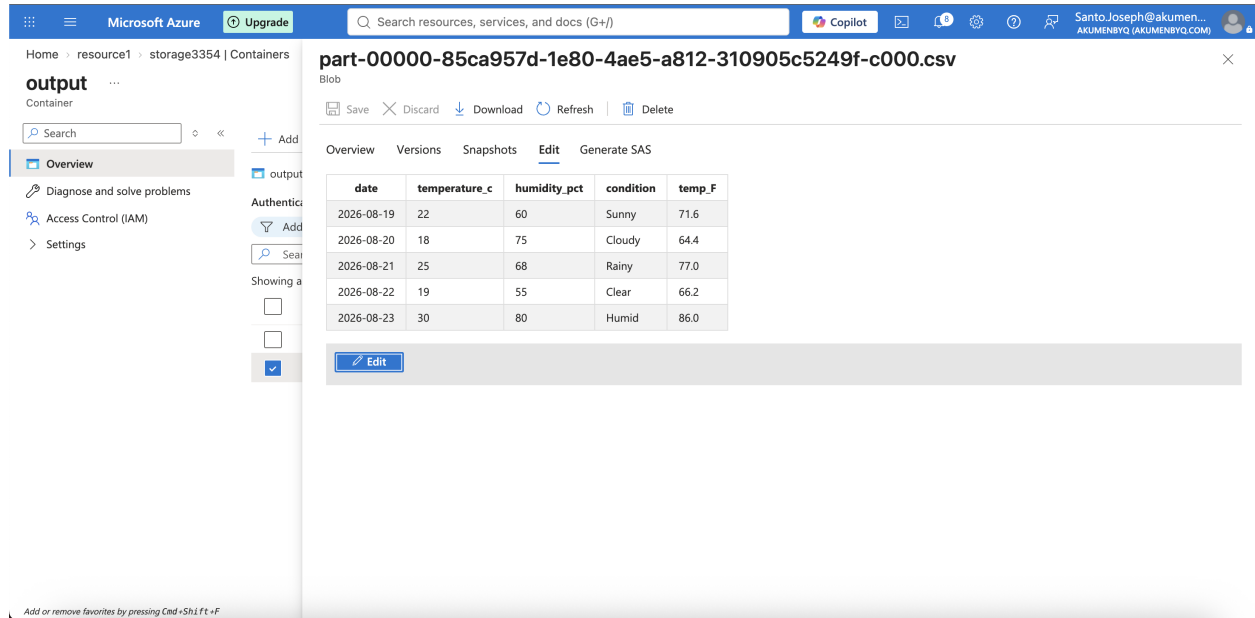
☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	_SUCCESS	24/08/2026, 15:39:03	Hot (Inferred)	Block blob	0	Available
☐	part-00000-85ca957d-1e80-4ae5-a812-310905c52...	24/08/2026, 15:39:02	Hot (Inferred)	Block blob	190 B	Available

part-00000-85ca957d-1e80-4ae5-a812-310905c5249f-c000.csv

Add or remove favorites by pressing Ctrl+Shift+F

Figure 19: Verify the output file in Blob Storage

Opening the generated CSV file in the storage Edit view confirmed the final dataset: **date, temperature_c, humidity_pct, condition, temp_F** — matching the Select transformation output exactly, with Celsius values correctly converted to Fahrenheit.



Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > resource1 > storage3354 | Containers

output ...

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

part-00000-85ca957d-1e80-4ae5-a812-310905c5249f-c000.csv

Blob

Save | Discard | Download | Refresh | Delete

Overview | Versions | Snapshots | **Edit** | Generate SAS

date	temperature_c	humidity_pct	condition	temp_F
2026-08-19	22	60	Sunny	71.6
2026-08-20	18	75	Cloudy	64.4
2026-08-21	25	68	Rainy	77.0
2026-08-22	19	55	Clear	66.2
2026-08-23	30	80	Humid	86.0

Edit

Add or remove favorites by pressing Cmd+Shift+F

Figure 20: Review the final output CSV